Wonjun Choi

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Fields

Econometrics Theory, Causal Inference, Machine Learning

Education

Emory University — **Atlanta, USA**
Ph.D. in Economics — *Spring 2027 (Expected)*

Korea University — **Seoul, Korea (South)**
Masters in Economics — *Spring 2021*
B.S. in Mathematics and Economics — *Spring 2018*

- Includes 2 years of military service.

Job Market Paper

A Machine Learning Approach to Affine Conditional Moment Restriction Models

This paper develops a general framework for estimation and inference in a class of affine conditional moment restriction models. Empirical applications of these models often involve a small number of endogenous regressors but a large set of instruments. Exploiting the affine structure, we leverage machine learning methods to develop an estimation procedure that is computationally fast and scales well to high dimensional instrumental variables. Using an orthogonalized minimum distance criterion, we characterize the requirement on the machine learners through their prediction errors, without restricting the analysis to a particular class of learners. Under this requirement, we develop a Wald statistic for hypotheses concerning smooth functionals. We also provide lower-level verification of our requirement for selected machine learning methods, as well as a Gaussian coupling of the estimated sieve coefficients useful for various inference problems. We examine the finite sample performance of our method through simulation studies and illustrate its use in an empirical application to migration in Tanzania.

Working In Progress

Partially Constrained Deep Learning with Qihui Chen and Zheng Fang (Draft available upon request)

This paper develops neural networks that enforce shape constraints by construction. In applications, particularly those with rich covariates, such constraints are often easier to justify for a subset of the inputs. Many existing methods, however, do not scale well to (even moderately) high dimensions. Neural networks, while well suited to such settings, face limitations: some constrain all inputs, whereas those allowing partial constraints may lack established universal approximation guarantees, suffer from severe overparameterization that impairs statistical inference, or both. We propose neural networks with a head that, in its basic form, evaluates the pointwise maximum over constraint-preserving combinations of outputs supplied by two branches: a constrained branch and a free branch. We also develop a variant that allows for the head to be deep. We formally show that these networks inherit universal approximation for partial constraints whenever the constrained branch has this property for the corresponding full constraints. Their relative parsimoniousness makes them attractive for statistical inference, which we illustrate via Monte Carlo simulations and empirical applications.

A Nonparametric Test of Supermodularity and Complementarity with Zhenting Sun and Zheng Fang

This paper develops a nonparametric test of supermodularity and complementarity.

Distributional Difference in Differences with Ka Yan Cheng, Florian Gunsilius

We propose a distributional difference-in-differences framework that includes quantile difference-in-differences and other methods as a special case.

Research Experience

Department of Economics, Emory University — Atlanta, GA
Research Assistant to Prof. Stefan Hoderlein — Spring 2026
Research Assistant: Prof. Zheng Fang — Fall 2025
Research Assistant: Prof. Maria Arbatskaya — Spring 2024, 2025
Research Assistant: Prof. Caroline Fohlin — Fall 2023
Research Assistant: Prof. Ian McCarthy — Fall 2022

Department of Political Science, Korea University — Seoul, Korea (South)
Research Assistant to Prof. Woochang Kang — Fall 2020

Korea Labor Institute — Sejong, Korea (South)
Research Assistant to Ph.D. Enhye Kwak — Spring 2020

Department of Economics, Korea University — Seoul, Korea (South)
Research Assistant to Prof. Jinyoung Kim — Fall 2019

Teaching Experience

Department of Economics, Emory University — Atlanta, GA
Teaching Assistant (ECON 524: Big Data Econometrics; Master) — Fall 2025
Instructor (ECON 220 Lab: Data Science for Economists; UG) — Spring 2025
Teaching Assistant (ECON 220: Data Science for Economists; UG) — Fall 2024

Department of Economics, Korea University — Seoul, Korea (South)
Teaching Assistant (Econometric Analysis; Graduate) — Spring 2021
Teaching Assistant (Microeconometrics; Graduate) — Spring 2020

Conference Presentations

Korean American Economic Association (KAEA) Job Market Conference — *November 2026*
Southern Economic Association (SEA) annual meeting *Huston* — *November 2026*

Skills

Software Python, R, MATLAB LaTeX

Language English (fluent), Korean (native)

References

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